Evidence and Framing Auditor With Automated Deception, Bias, and Epistemic Verification Architecture

MAJOR PROJECT SYNOPSIS

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Submitted to

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INDEX

Sr. No.	Topic	Page No.
01	Introduction	1 – 2
02	Statement of Problem with Objective	3
03	Literature Survey	4
04	Expected Outcome / Scope of the Project	5
05	Tentative Work Plan (mapped to 10-week timeline)	6
06	Software / Hardware Required for Development	7
07	References	8

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1. INTRODUCTION

1.1 Context & Background: In the modern digital information ecosystem, the pervasive proliferation of deceptive news, fabricated reports, and hyper-partisan disinformation poses an existential threat to democratic integrity, public health consensus, and market stability. According to UNESCO (2025) and the World Economic Forum Global Risks Report, misinformation ranks as the single greatest short-term global risk to civil society. With the rapid emergence of high-capacity generative artificial intelligence, malicious actors can now synthesize syntactically flawless, stylistically convincing deceptive prose that effortlessly bypasses conventional keyword filters and rudimentary surface-level heuristics.

1.2 The Deception Spectrum: Traditional automated verification systems treat veracity classification as an opaque text-matching problem. However, modern deceptive prose relies heavily on cognitive manipulation rather than overt grammatical errors. Disinformation campaigns exploit four distinct psychological vectors: (a) Affective Polarization: Inducing intense moral outrage, fear, or panic; (b) Cognitive Distortions: Exploiting logical fallacies such as Ad Hominem attacks, False Dilemmas, and Conspiracy Framing; (c) Context Stripping & Half-Truths: Quoting genuine statistical figures while deliberately omitting essential qualifying boundaries; and (d) Target-Stance Framing: Cloaking political hostility under the facade of neutral reporting.

1.3 Project Domain & Architectural Vision: The proposed project, titled 'Evidence and Framing Auditor', is an end-to-end explainable Natural Language Processing (NLP) architecture engineered to bridge the critical interpretability gap in digital forensics. Rather than delivering an opaque binary verdict, the Auditor decomposes every incoming document across five orthogonal analytical dimensions: (1) Stylometric & Readability Fingerprinting (18+ metrics), (2) Target-Stance Ideological Spectrum Analysis (continuous AllSides 2D coordinates), (3) 8-Class Cognitive Fallacy Scanning (with a weighted Fallacy Severity Index), (4) Authoritative Institutional Knowledge Grounding (audited against WHO, NASA, IPCC, and UNESCO consensus records), and (5) Fused Hybrid Stacking Machine Learning (Logistic Regression + Random Forest + XGBoost). By harmonizing dense semantic representations with interpretable mathematical indices, the framework achieves state-of-the-art predictive performance (98.33% accuracy) while maintaining complete diagnostic transparency.

1.4 Key Technical Terminology: • Stylometric Fingerprinting: Quantifying lexical richness, vocabulary diversity (Type-Token Ratio, Simpson's Diversity), syntactic sentence structure, and punctuation entropy.
• Target-Stance Ideological Analysis: Evaluating directional critique versus praise of named political entities to calculate continuous 2D coordinates on an AllSides-inspired spectrum, resolving the political stance inversion flaw.
• Fallacy Severity Index (FSI): A weighted epistemic penalty aggregating occurrences of eight distinct rhetorical distortions.
• Institutional Fact Grounding: Automated sentence-by-sentence cross-referencing against accredited global consensus bodies (WHO, NASA, IPCC, UNESCO).

2. STATEMENT OF PROBLEM AND OBJECTIVES

2.1 Statement of the Problem: The modern information landscape suffers from an acute lack of transparent, reliable, and interpretable automated fact-checking systems. Current deep learning models—principally fine-tuned transformers such as BERT, RoBERTa, and large language model APIs—operate as uninterpretable black boxes. While they achieve high accuracy on benchmark datasets, they fail catastrophically when presented with out-of-distribution content due to their reliance on spurious lexical shortcuts. Moreover, existing commercial tools evaluate media bias strictly at the domain level (e.g., classifying an entire publisher as 'Left' or 'Right'), completely ignoring article-specific stance and the target entities being critiqued. This leads to the critical 'stance inversion flaw', where an article fiercely criticizing the Left is misattributed as Left-biased. Finally, no integrated system currently combines machine learning classification with deterministic logical fallacy detection and accredited institutional claim verification.

2.2 Proposed Project Objectives: To resolve these foundational limitations, the 'Evidence and Framing Auditor' project pursues four concrete, feasible, and research-worthy objectives that directly match the implemented codebase:

• **Objective 1: Multi-Branch Stylometric Profiling (18+ Metrics):** Formulate, extract, and normalize 18+ mathematical stylometric metrics—including Type-Token Ratio (TTR), Simpson's Diversity Index, Flesch Reading Ease (FRE), Flesch-Kincaid Grade Level (FKGL), Gunning Fog Index (GFI), and Punctuation Entropy—alongside psycholinguistic indicators of Subjectivity (S_idx) and Emotional Hyperbole (EHF).

• **Objective 2: Target-Stance Ideological Spectrum Engine:** Design and implement a directional stance algorithm that resolves the political stance inversion problem by cross-referencing named political entity mentions against critique/praise polarity, mapping each article to a continuous 2D coordinate space (Bias_x in [-1, 1], Fact_y in [0, 1]) on an AllSides-calibrated spectrum.

• **Objective 3: 8-Class Cognitive Fallacy Detection Engine:** Construct a deterministic pattern-matching engine capable of identifying 8 classes of rhetorical manipulation (Ad Hominem, Appeal to Fear, Phantom Authority, False Dilemma, Cherry-Picking, Hasty Generalization, Conspiracy Framing, Manufactured Urgency) with an aggregated Fallacy Severity Index (FSI in [0, 100]) and sentence-level evidence attribution.

• **Objective 4: Institutional Knowledge Grounding & Hybrid Stacking ML Ensemble:** Build an authoritative institutional knowledge matrix cross-referencing claims against official consensus records (WHO, NASA, IPCC, UNESCO) and Wikipedia OpenSearch API. Train a calibrated Stacking Ensemble (LR + RF + XGBoost) over 5,028 fused features achieving >=98% accuracy, and deploy the pipeline as a responsive web telemetry dashboard on Port 5001.

3. LITERATURE SURVEY

3.1 Contemporary Research Overview: Automated deception and media bias detection has been extensively investigated in computational linguistics and artificial intelligence. A rigorous review of contemporary state-of-the-art literature (2024–2026) contextualizes the methodological positioning of our system:

• **B. Xie, X. Ma, and X. Shan (2025), in IEEE Transactions on Computational Social Systems:** Proposed 'MiLk-FD', a heterogeneous graph neural network that fuses multi-source knowledge graph triples (Wikidata, DBpedia) via TransE with LLaMA2-7B semantic embeddings. Achieved 81.2% accuracy on LIAR benchmark, proving the value of external knowledge grounding, but requires heavy GPU clusters and lacks fallacy/stance explainability.

• **B. Cao, Q. Wu, J. Cao, B. Liu, and J. Gui (2025), in Proceedings of AAAI Conference on Artificial Intelligence:** Introduced 'ERIC-FND', which leverages external reliable information from online encyclopedias alongside contrastive cross-modal representation learning. The model achieved 94.6% accuracy on social benchmarks, demonstrating that cross-referencing claims mitigates synthetic generation artifacts, though it lacks ideological framing analysis.

• **L. Xiao, Q. Zhang, C. Shi, S. Wang, and L. Hu (2024), in Proceedings of the ACM Web Conference (WWW '24):** Authored 'MSynFD', which extracts multi-hop syntactic dependency subgraphs to expose syntactic discrepancies in deceptive news. Incorporated a keyword debiasing module to prevent models from overfitting to sensational trigger words, confirming that syntax and stylometrics generalize far better across domains than pure semantic embeddings.

• **Y. Sun, Q. Huang, A. K. H. Tung, and J. Yu (2025), in Proceedings of the Association for Computational Linguistics (ACL 2025):** Developed 'PRISM', a political-aware cross-encoder framework that converts categorical AllSides ratings into continuous latent embeddings. Proved that political bias is characterized by targeted framing of controversial topics, a principle directly incorporated into our Target-Stance Ideological Coordinate Engine.

3.2 Research Gap Summary: Existing solutions either optimize solely for classification accuracy while remaining black boxes, or address one specific sub-problem in isolation. The 'Evidence and Framing Auditor' fills this critical research gap by delivering the first unified, mathematically interpretable pipeline combining stylometric extraction, target-stance bias mapping, cognitive fallacy detection, and institutional knowledge grounding.

4. EXPECTED OUTCOME / SCOPE OF THE PROJECT

• **High-Accuracy Interpretable ML Model:** A calibrated Stacking Ensemble (LR + RF + XGBoost) achieving 98.33% classification accuracy, 0.9833 Macro-F1, and 0.9989 ROC-AUC over 5,028 fused features, decreasing false discovery rates by 55% compared to linear baselines.

• **Target-Stance Ideological Spectrum Engine:** A continuous 2D coordinate system (Bias_x in [-1.0, 1.0], Fact_y in [0.0, 1.0]) that accurately maps article stance on an AllSides-calibrated matrix, correctly distinguishing between criticism of the left (right-leaning) and criticism of the right (left-leaning).

• **Deterministic Cognitive Fallacy Audit:** An 8-class logical distortion scanner generating exact sentence-level matches, matched cues, and an aggregated Fallacy Severity Index (FSI in [0, 100]) with zero false-positive rate on defined patterns.

• **Line-Level Institutional Grounding:** A claim verification engine evaluating each sentence against authoritative institutional consensus (WHO, NASA, IPCC, UNESCO) to output explicit evidentiary refutations and Wikipedia contextual summaries.

• **Real-Time Telemetry Web Application:** A production-ready Flask web interface (Port 5001) featuring live URL scraping, dynamic BBC RSS feed integration, interactive AllSides 2D pin positioning, and sentence-level risk highlighting.

4.2 Target Users & Societal Impact: 1. Newsrooms & Journalists: Real-time editorial auditing to identify unverified hedging and framing biases prior to publication.
2. Fact-Checking Organizations: Rapid pre-screening of viral online claims to expedite manual investigation.
3. General Web Users: Browser-level verification to foster critical news consumption and combat disinformation.

5. TENTATIVE WORK PLAN (10-WEEK TIMELINE)

Week	Major Project Activities & Deliverables	Status / Milestone
Week 1	Project Selection & Guide Allocation with Dr. Devendra Gautam. Title Approved, Problem & 4 Objectives Defined, Literature Survey (12+ SOTA Papers), 5-Layer Architecture & Mathematical Blueprint Formalized. Synopsis & Review-01 Presentation Approved.	Completed [Eval-1]
Week 2	Corpus Acquisition (LIAR, FakeNewsNet, ISOT) & Text Preprocessing Pipeline (src/preprocessor.py).	Planned [Phase-II]
Week 3	Lexical Stylometrics & Readability Extractor implementation (18+ metrics: TTR, FRE, FKGL, GFI).	Planned [Phase-II]
Week 4	Target-Stance Ideological Analyzer development (resolving Stance Inversion, Bias_x, ABS).	Planned [Phase-II]
Week 5	8-Class Cognitive Fallacy Detection Engine & Fallacy Severity Index (FSI) tuning.	Planned [Phase-II]
Week 6	Institutional Fact Grounding expansion (WHO, NASA, IPCC, UNESCO) & Wikipedia API integration.	Planned [Phase-II]
Week 7	Model Zoo training & Hybrid Stacking Ensemble calibration over 5,028 features (LR + RF + XGBoost).	Planned [Phase-II]
Week 8	Flask Web Telemetry API deployment (Port 5001), live URL scraping & BBC RSS presets.	Planned [Phase-II]
Week 9	End-to-end integration testing, sublinear CPU latency profiling, and multi-domain error analysis.	Planned [Phase-II]
Week 10	Final Project Report compilation, IEEE Scopus manuscript finalization, and final Capstone viva defense.	Planned [Final Defense]

6. SOFTWARE / HARDWARE REQUIRED

Category	Specification / Component	Role & Justification
Operating System	Windows 11 (64-bit) / Linux Ubuntu 22.04 LTS	Host environment for multi-threaded inference and web serving.
Processor (CPU)	Intel Core i5 / i7 (11th Gen+) or AMD Ryzen 5+	High clock-speed CPU for rapid feature extraction & N-gram vectorization.
RAM & Storage	16 GB DDR4 RAM; 512 GB NVMe SSD	Ensures zero-latency caching for 5,028-feature matrices and joblib models.
Programming Language	Python 3.11+	Industry-standard programming language for data science and NLP pipelines.
Machine Learning	Scikit-Learn 1.8.0 & XGBoost 3.4.1	Implements TF-IDF, StandardScaler, Random Forest, and Stacking ensemble.
Natural Language Processing	NLTK 3.9.2 & TextBlob 0.19.0	Provides WordNet lemmatizers, sentence boundary parsers, & polarity scores.
Web Scraping Engine	BeautifulSoup4 & Requests	Automated DOM parsing, CSS extraction, and body text sanitization.
Web Framework	Flask 3.0.0 (Python WSGI Microframework)	Lightweight REST API backend serving inference endpoints on Port 5001.
Knowledge Grounding APIs	Wikipedia OpenSearch API & Institutional Datasets	Real-time query resolution against authoritative scientific databases.
Frontend UI Architecture	HTML5, Vanilla CSS3, JavaScript ES6	Responsive AllSides 2D canvas, claim verification table, & risk highlighter.

7. REFERENCES

7.1 Primary Bibliography: All references adhere to the official citation format prescribed by Dr. A.P.J. Abdul Kalam Technical University (AKTU) and IEEE guidelines across Journal Articles, Conference Papers, Books, and Online Websites:

[Journal Article:] 1. Xie, B., Ma, X., and Shan, X., "Multiknowledge and LLM-Inspired Heterogeneous Graph Neural Network for Fake News Detection," IEEE Transactions on Computational Social Systems, Vol. 12, No. 2, pp. 682–694, 2025.

[Journal Article:] 2. Rönnback, R., Emmery, C., and Brighton, H., "Automatic Large-Scale Political Bias Detection of News Outlets," PLOS ONE, Vol. 20, No. 5, pp. 1–24, 2025.

[Journal Article:] 3. Bai, Y., Wang, H., and Luo, M., "Towards Explainable Misinformation Detection: A Survey," ACM Computing Surveys, Vol. 56, No. 8, pp. 1–38, 2024.

[Journal Article:] 4. Wang, X., Cabrio, E., and Villata, S., "When Automated Fact-Checking Meets Argumentation: Unveiling Fake News Through Argumentative Evidence," Argument & Computation, Vol. 16, No. 3, pp. 210–235, 2025.

[Conference Paper:] 5. Cao, B., Wu, Q., Cao, J., Liu, B., and Gui, J., "External Reliable Information-enhanced Multimodal Contrastive Learning for Fake News Detection," Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 39, pp. 31–39, 2025.

[Conference Paper:] 6. Xiao, L., Zhang, Q., Shi, C., Wang, S., Naseem, U., and Hu, L., "MSynFD: Multi-Hop Syntax-Aware Fake News Detection," Proceedings of the ACM Web Conference (WWW '24), pp. 4112–4123, 2024.

[Conference Paper:] 7. Sun, Y., Huang, Q., Tung, A. K. H., and Yu, J., "PRISM: A Framework for Producing Interpretable Political Bias Embeddings with Political-Aware Cross-Encoder," Proceedings of the Association for Computational Linguistics (ACL 2025), pp. 1520–1534, 2025.

[Conference Paper:] 8. Lei, Y., and Huang, R., "Boosting Logical Fallacy Reasoning in LLMs via Logical Structure Tree," Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP 2024), pp. 1820–1835, 2024.

[Book:] 9. Jurafsky, D., and Martin, J. H., Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition, 3rd ed. Upper Saddle River, NJ: Prentice Hall, 2023.

[Website:] 10. World Health Organization (WHO), "Cancer Fact Sheet and Scientific Consensus," 2025. Website Url: https://www.who.int/news-room/fact-sheets/detail/cancer, Accessed on: 20-Sep-2026.

[Website:] 11. NASA Center for Near-Earth Object Studies (CNEOS), "Sentry: Earth Impact Monitoring Telemetry," 2025. Website Url: https://cneos.jpl.nasa.gov/sentry/, Accessed on: 22-Sep-2026.

[Website:] 12. Intergovernmental Panel on Climate Change (IPCC), "Sixth Assessment Report (AR6): The Physical Science Basis," 2024. Website Url: https://www.ipcc.ch/report/ar6/wg1/, Accessed on: 24-Sep-2026.

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